Introduction

In today’s fast-paced IT industry, professionals are expected to continuously learn, upgrade their technical skills, and apply concepts in real-world scenarios. However, busy schedules, remote locations, and limited availability of convenient time slots often become major barriers to hands-on implementation. This challenge extends to students as well, especially in IT education, where technology evolves rapidly.

Fortunately, technology itself provides a solution!

At **Aptech**, we are committed to leveraging technology in our training model. To revolutionize the way students, learn and apply concepts, we introduce **eProject**—a **structured, and interactive learning environment** designed to simulate real-world project implementation.

What is eProject?

eProject is a step-by-step, guided learning experience that mirrors classroom and lab-based training, allowing students to:

☑ **Practice progressively** using a laddered approach.
☑ **Build robust applications** from the ground up.
☑ **Utilize essential utilities** in user-designed applications.
☑ **Transform individual programs** into a unified, complete application.
☑ **Implement concepts in phases** for deeper understanding.
☑ **Enhance skills and add value** to their expertise.
☑ **Work on real-life projects** with practical relevance.
☑ **Develop complex and useful applications** based on real-world scenarios.
☑ **Receive mentoring and guidance** via email support.

How It Works

Students are required to complete their **eProject**, document their progress, and submit the source code within the allotted timeframe to the **eProjects Team**.

We look forward to your enthusiastic participation and a successful learning journey with **eProject**!

Objectives of the project

The primary objective of this program is to provide a **hands-on experience** in working on real-life projects. These applications serve as building blocks, helping students develop **larger and more robust applications**.

Rather than focusing on teaching specific software, this program aims to **present real-world scenarios** that enable students to create **practical applications** using the available tools.

Before starting the project, students are encouraged to **revise relevant topics** to ensure a strong foundation.

Guidelines for Execution

- Projects should be completed during **lab sessions**, with faculty assistance if needed.
- A **clear understanding of the subject matter** is essential for successful implementation.
- For any queries related to the project or its objectives, students should reach out to the **eProjects Team**.

We look forward to seeing your innovative applications come to life!

Problem Statement

1. Introduction

1.1 Background and Necessity for the Web Application

Access to healthcare professionals can be difficult, especially for people living in remote areas or for those with limited mobility. Traditional healthcare systems may involve long wait times for appointments, geographical constraints, and inadequate access to specialized care. Moreover, maintaining a patient's health history across various visits and institutions can often lead to fragmented data, which hinders efficient treatment. A comprehensive platform that allows for easy scheduling of virtual consultations, seamless management of health records, and basic AI-driven diagnoses could significantly improve access to healthcare, reduce wait times, and streamline the entire healthcare process.

The **Online Health Consultation Portal** aims to provide an efficient, secure, and accessible way for patients to receive medical consultations, store health records, and use AI for preliminary diagnosis. By reducing barriers to access and providing a more streamlined healthcare experience, this platform has the potential to revolutionize the way people engage with healthcare services.

1.2 Proposed Solution

The **Online Health Consultation Portal** is a secure web-based platform that allows patients to book virtual consultations with doctors, maintain and access their health records, and receive basic AI-driven diagnostics. This solution aims to enhance access to healthcare, especially for those who are unable to visit a healthcare facility physically. By providing tools for managing health records digitally and utilizing AI for preliminary diagnoses, the platform will make healthcare more efficient, accessible, and patient-centric.

1.3 Purpose of the Document

This document outlines the functional and non-functional requirements, technical specifications, and constraints of the Online Health Consultation Portal (WebApp). It serves as a guideline for developers, designers, and stakeholders involved in the project.

1.4 Constraints

The development of the Online Health Consultation Portal Web application must adhere to several constraints to ensure its successful implementation and operation. Technically, the application must be compatible with major Web browsers and responsive across various devices. You may encounter constraints related to data storage, data synchronization, and backup procedures. The usage of various images and videos may be subject to licensing agreements and copyright restrictions. It is important to understand and comply with these constraints to avoid legal issues.

2. Functional Requirements

The Web application will be designed with a set of forms/pages with menus representing choice of activities to be performed.

Following are the functional requirements of the application:

1. **Virtual Consultation Scheduling**:
    - Patients can easily browse a list of available doctors (general practitioners, specialists) and schedule appointments based on their preferred time and doctor availability.
    - Integration with a calendar system to manage appointment slots and send automatic reminders for upcoming consultations.
    - Option to choose the type of consultation (e.g., video call, audio call, chat) based on the patient's preferences and connectivity.
    - Ability to reschedule or cancel appointments with prior notice.
2. **Doctor Profiles**:
    - Detailed doctor profiles including their qualifications, specialties, years of experience, and patient ratings and reviews.
    - Filtering options to help patients select doctors based on specialty, availability, language spoken, and location.
3. **Patient Health Records Management**:
    - A secure, cloud-based platform to store and manage health records, including medical history, prescriptions, lab results, imaging reports, and past consultations.
    - The ability to upload documents and images (e.g., scans, X-rays) for doctor review.
    - Easy access to historical health data, allowing patients and doctors to track treatment progress over time.
    - Consent-based sharing of health records between different healthcare providers if needed.
4. **Video Consultation**:
    - A secure video calling feature integrated into the platform to facilitate real-time consultations.
    - Screen sharing for doctors to explain diagnoses, treatment plans, and share documents or images.

5. **Prescription and Follow-Up**:
   - Doctors can write prescriptions during consultations and send them directly to the patient via the platform.
   - Patients can view their prescriptions and receive reminders for medication refills or follow-up appointments.
   - A follow-up feature to schedule additional appointments or check-ups if needed.
6. **Chat and Text-Based Communication**:
   - Secure messaging system for text-based consultations, where patients can ask questions and receive answers from their doctor in a timely manner.
   - Option for asynchronous communication for less urgent matters.
7. **Secure Payment Integration (Optional)**:
   - In-app payment options to process payments for consultations securely. Show only invoice/bills generated. No Payment Gateways to be used.
   - Integration with insurance providers to enable billing and payments, or offer financial assistance for consultations. Only details of Insurance can be collected from the user.
8. **Health Reminders and Notifications**:
   - Automated reminders for scheduled consultations, medication refills, vaccinations, health screenings, and follow-up visits.
   - Notifications for new test results or updates on previously scheduled appointments.
9. **Admin Dashboard for Healthcare Providers**:
   - An admin dashboard for healthcare providers to manage consultations, track patient progress, and view diagnostic tools.
   - Ability to review, approve, and update patient health records.

3. Technical Considerations

Hardware

- Intel Core i5 Processor or higher
- 8 GB RAM or above
- Color SVGA
- 500 GB Hard Disk space
- Mouse
- Keyboard

Software
Technologies to be used:

Frontend: HTML5, CSS3, Bootstrap, ReactJS 18 or higher/AngularJS/Angular 9 or higher, JavaScript, jQuery, and XML

Backend: Java 16 or higher (with Apache NetBeans or Eclipse), Java EE 7 or higher/Jakarta EE 9 or higher or C# 7.2 with ASP.NET MVC and ASP.NET MVC Core (optional), Visual Studio 2019 or higher or PHP 7.2 or higher with Laravel Framework or Python 3.0 or higher with Flask or Django

Database: MySQL 8.0 or higher/SQL Server 2019 or higher

For local hosting: XAMPP latest version

4. Non-Functional Requirements

There are several non-functional requirements that should be fulfilled by the application. These include:

- **Safe to use:** The application should not result in any malicious downloads or unnecessary file downloads.
- **Accessibility:** The application should have clear and legible fonts, user-interface elements, and navigation elements.
- **User-friendliness:** The application should be easy to navigate with clear menus and other elements and easy to understand.
- **Operability:** The application should be reliable and efficient.
- **Performance:** The application should demonstrate high value of performance through speed and throughput. In simple terms, the application should have minimal load time and smooth page redirection.
- **Scalability:** The application architecture and infrastructure should be designed to handle increasing user traffic, data storage, and feature expansions.
- **Security:** The application should implement adequate security measures such as authentication. For example, only registered users can access certain features.
- **Availability:** The application should be available 24/7 with minimum downtime.
- **Compatibility:** The application should be compatible with latest browsers and various devices.

5. Project Deliverables

You will require to design and build the project and submit it along with a complete project report that includes:

- Problem Definition
- Design specifications
- Diagrams such as flowcharts for various activities, Data Flow Diagrams, and so on
- Database Design
- Test Data Used in the Project
- Project Installation Instructions (if any)
- User Credentials for all types of users with passwords

Documentation is considered as a very important part of the project. Ensure that documentation is complete and comprehensive. The consolidated project will be submitted as a zip file with a ReadMe.doc file listing assumptions (if any) made at your end and SQL scripts files (.sql) containing database and table definitions.

Documentation should not contain any source code.

Preferably, host the working Web application on a site and share the URL for evaluation. In addition, **you must submit a video clip** (mp4 file) showing the actual working of the application.

Over and above the given specifications, you can apply your creativity and logic to improve the system.

Sitemap: To understand the flow of **Online Health Consultation Portal** Web Application, you will have to create a Sitemap and add it to the home page of your application.